

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 4

Team Members :

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1. Problem Title :

Student Attendance Management System

2. Problem Description:

Imagine the start of every class. The professor pulls out a stack of papers or a huge register and starts calling out names one by one. It's slow, takes up valuable lecture time, and it's easy to make mistakes—maybe you mark the wrong person absent, or can't read someone's handwriting. Later, if a student wants to know their attendance percentage, it's a manual chore to count it all up.

Our goal is to fix this frustrating, old-school process. We want to build a simple, digital system where faculty can mark attendance in seconds on a computer or tablet. This data will be stored securely and automatically updated. Students can then log in anytime to see exactly where they stand, without having to chase anyone for the information. For administrators, this system will provide a clear, bird's-eye view of attendance trends across different courses, helping them identify students who might be at risk. In short, we're replacing a tedious, error-prone chore with a fast, accurate, and transparent tool for everyone.

3. Identified Entities (Classes)

Class Name	Description
Student	This represents each student using the system. Their main goal is to log in and quickly check their own attendance for their subjects.
Faculty	This is for the professors or teachers. Their main job is to conduct a class and easily mark who is present and who is absent.
Admin	This is the system overseer. They don't mark attendance themselves, but they manage the big picture, like adding new subjects or users.
Subject	This represents a specific course, like "Object Oriented Programming." Attendance is tracked for a subject.
Session	A single class on a specific day is a "Session." For example, the OOP class on Monday at 9 AM is one session. Attendance is marked <i>per session</i> .
AttendanceRecord	This is the actual link between a student and a session. It's a single entry that says, "Student X was Present/Absent for Session Y."

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Student	- studentID (String) - name (String) - rollNumber (String) - email (String) - password (String)	- login(username, password) - logout() - viewAttendance(subjectCode) - getAttendancePercentage(subjectCode)
Faculty	-facultyID(String) -name (String) - email (String) - password (String) - department (String)	- login(username, password) - logout() -markAttendance(sessionID, studentAttendanceList) - viewStudentList(subjectCode) - generateAttendanceReport(subjectCode, fromDate, toDate)
Admin	- adminID (String) - name (String) - email (String) - password (String)	- login(username, password) - logout() - addSubject(subjectCode, subjectName) - removeSubject(subjectCode) -addNewUser(userDetails) - viewOverallAttendanceTrends()
Subject	-subjectCode (String) - subjectName (String) -facultyID (String) -credits(int)	- getSubjectDetails() - getEnrolledStudents() -calculateOverallAttendance()

Session	- sessionID (String/UUID) - subjectCode (String) - date (Date) - startTime (Time) - endTime (Time)	- createSession(subjectCode, date) - getAttendanceRecords() - markStudentAttendance(studentID, status)
AttendanceRecord	- recordID (String/UUID) - studentID (String) - sessionID (String) - status (Enum: PRESENT, ABSENT, LATE, EXCUSED) -timestamp (DateTime)	- updateStatus(newStatus) - getStatus()